RESEARCH & INNOVATION

Case Study Options

ELIGIBLE DISCIPLINES

Open to pre-final and final-year students with Bachelor's degree or a Master's degree in relevant disciplines, including Computational Chemistry, Chemical Engineering, Chemistry, Materials Science, Polymer Science & Technology, Surface/Polymer Technology, Materials Engineering, Green Technology, Pharmaceutics, Chemical Technology, Pharmaceutical Chemistry, Pharmacology, Pharmacognosy, Biosciences, Bioengineering, Biochemical Engineering, Biotechnology, Biological Science, Natural Product Chemistry, Computational Physics, Pharmaceutical Analysis, Bioinformatics, Computational Biology, Applied Mechanics, Pharma Tech, Molecular Dynamics, Predictive Formulation, Computational Chemistry & Data Science.



OPTION 01

Engineering Next-Generation Bio-Sourced Mineral oil Replacement

CONTEXT

Mineral oil is backbone of cosmetic formulations across skin care, hair care, hair coloration. Sustainability priority is replacing petroleum-derived hydrocarbons.

Nature-derived lipids and plant-based vegetable oils often fail under pressure: thermal stability, oxidation, maintaining integrity in extreme environments.

Advanced hair colorations/treatments span pH 2 to 12; natural oils can break down, separate, alter viscosity.

India Biodiversity Act adds regulatory complexity.

CHALLENGE

Investigate scalable, high-performance bio-sourced alternative to mineral oil like Biosourced Alkane or others that can withstand extreme chemical environments without compromising sensory appeal/stability.

KEY DELIVERABLES

- Identify/investigate plant-derived, microbial, upcycled emollients/bio-alkanes; evaluate viscosity, spreadability, oxidative stability vs mineral oil.
- Propose physicochemical strategies such as smart emulsification, hybrid structural matrices, molecular tuning to keep bio-sourced oils stable across pH 2–12, preventing phase separation, coalescence, rheological breakdown.
- Prove commercial viability: green chemistry indicators, LCA, sourcing constraints incl. India Biodiversity Act, large-scale manufacturing feasibility.

OPTION 02

Longevity and Future Aging for GenZ in the Era of Climate Change and Pollution

CONTEXT

For GenZ India, aging is holistic skin health/wellness/resilience, not just wrinkles.

Rising temperatures, extreme weather, urban pollution raise anxiety about accelerated aging/environmental stressors.

R&I should develop solutions for longevity, cellular defense, emotional reassurance, adaptive skincare, hybrid protective formulations, digital diagnostics tracking/preventing/reversing early invisible climate-induced aging signs like barrier degradation, oxidative stress, dullness.

CHALLENGE

Step into L'Oréal researcher role and develop groundbreaking cosmetic product, devices, digital solutions for India; combine scientific innovation with Indian consumer understanding; target L'Oréal Paris, Garnier, CeraVe.

CONTINUED

Longevity and Future Aging for GenZ in the Era of Climate Change and Pollution

KEY THOUGHT PROCESS

01

Common Indian Skincare Challenges

Skin tone, oily skin, acne, sun damage, pollution exposure, or other.

02

Climate & Lifestyle

Hot/humid, varying pollution, impact on formulation/usage.

03

Cultural Nuances & Preferences

Grooming habits, ingredient/formats, attitudes.

04

Innovation & Technology

Cutting-edge ingredients, technologies, formulations; beyond traditional.

05

Sustainability & Ethical Sourcing

Environmental impact, sustainable ingredients/packaging.

06

Marketing & Positioning

Target Indian consumers, messaging/channels.

DELIVERABLES

01

Product Concept

Detailed description of cosmetic product/device/digital solution, name, audience, ingredients, texture, format, technology, benefits.

02

Scientific Rationale

Scientific basis, mechanisms of action, formulation/technology rationale; sound scientific principles.

03

Marketing Strategy

Target segmentation, key messaging, channels.

04

Sustainability Plan

Sustainable/ethical practices.

OPTION 03

Green & Sustainable Permanent Coloration

Engineering Bio-Derived Anthocyanin Cream Formulations for Wash-Fast Hair Deposition

CONTEXT

Permanent hair coloring relies heavily on synthetic oxidative dye precursors and aggressive alkaline swelling agents. Transitioning to green, bio-derived alternatives—specifically plant-derived anthocyanins—presents severe chemical hurdles. Anthocyanins are inherently water-soluble, structurally sensitive to pH shifts, and prone to rapid leaching and fading during rinsing. Furthermore, formulating these natural pigments into a high-performance cream emulsion (avoiding messy powders like henna or indigo) requires precise control over rheology, mass transfer, and substantive deposition onto the keratin protein structure without relying on traditional synthetic developers.

CHALLENGE

Identify next-generation sustainable coloration systems that deliver vibrant, long-lasting shades through bio-sourced chemistry. This requires designing novel supramolecular delivery strategies, optimizing emulsion rheology for cream matrices, and establishing green chemical processes that ensure high wash-fastness from root to tip.

CONTINUED

Green & Sustainable Permanent Coloration

KEY THOUGHT PROCESS & DELIVERABLES

01

Investigate Pigment-Fiber Interactions & Color Permanence

Analyze the molecular and supramolecular interactions (e.g., hydrogen bonding, electrostatic attraction, or complexation) between anthocyanin chromophores and the keratin protein matrix to achieve wash-fast, long-lasting color retention.

02

Design Advanced Cream Emulsion Systems

Propose novel physicochemical strategies for formulating bio-derived pigments into high-viscosity cream matrices, focusing on rheological profiling, phase behavior, and mass transfer to ensure uniform spreadability and deep fiber deposition.

03

Evaluate Green Chemistry & Scale-Up Constraints

Assess the chemical engineering and industrial scale-up challenges of bio-sourced colorant sourcing, mass production of sustainable cream bases, and adherence to green chemistry metrics (atom economy, life-cycle analysis, biodegradability).